LIFESTYLE-BASED MENTAL HEALTH WELLBEING PREDICTION

DEEP LEARNING & POWER BI DASHBOARD PROJECT

PROJECT REPORT

BY

ABHIJITH J

DATA SCIENTIST

UNDER THE GUIDANCE & SUPERVISION OF

Ms. AMRITHA MR

Palakkad, Kerala, 678702

TECHOLAS TECHNOLOGIES

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1. INTRODUCTION

Mental health wellbeing has become a crucial aspect of overall human health in modern society due to increasing stress levels, unhealthy lifestyles, and lack of proper sleep patterns.

Various lifestyle factors significantly affect mental wellbeing such as:

- Sleep duration
- Stress levels
- Physical activity
- Screen time
- Work-life balance
- Diet habits

These factors directly influence an individual's psychological health and overall wellbeing.

With the advancement of Artificial Intelligence and Deep Learning, predictive models can be developed to analyze lifestyle patterns and estimate mental health wellbeing. This project focuses on building an Artificial Neural Network (ANN) model to predict whether a person’s wellbeing is **Poor, Average, or Good** based on lifestyle factors.

2. ABSTRACT

This project focuses on predicting mental health wellbeing using an Artificial Neural Network (ANN) classification model. Mental wellbeing is influenced by various lifestyle factors such as sleep duration, stress levels, physical activity, and screen time.

The objective of this project is to develop a predictive model that classifies individuals into three categories: Poor, Average, and Good wellbeing.

The project involves data preprocessing, exploratory data analysis (EDA), ANN model development, hyperparameter tuning, and performance evaluation using classification metrics. The model is trained to identify patterns between lifestyle habits and mental health outcomes.

Additionally, a Streamlit web application is developed to provide real-time predictions based on user input, making the system interactive and user-friendly.

3. OBJECTIVES

- Build an ANN classification model for wellbeing prediction
- Analyse relationship between lifestyle factors and mental health
- Optimize ANN model performance
- Evaluate model using classification metrics
- Evaluate model performance using regression metrics
- Develop Streamlit app for prediction

4. METHODOLOGY

Step 1: Data Collection:

- Dataset sourced from Kaggle - Mental Health and Lifestyle Research dataset

Step 2: Data Preprocessing:

- Handled missing values
- Encoded categorical variables
- Feature scaling using StandardScaler
- Target variable encoded (Poor, Average, Good)

Step 3: Exploratory Data Analysis (EDA):

EDA performed to understand:

- Wellbeing Distribution (Bar chart)
- Wellbeing Category (Pie chart)
- Sleep vs Wellbeing (Bar chart)
- Stress Level vs Wellbeing (Bar chart)
- Correlation Heatmap of Lifestyle Factors

Step 4: Model Building:

An Artificial Neural Network (ANN) model was developed for mental wellbeing level prediction.

The neural network consists of:

- Input layer representing lifestyle features
- Hidden dense layers
- Activation functions (ReLU)
- Dropout layers for overfitting prevention
- Output layer for Classification

Step 5: Model Evaluation:

The ANN model was evaluated using classification metrics such as Accuracy Score, Confusion Matrix, Precision,Recall, F1 Score.

Step 6: Dashboard Development:

Interactive Power BI dashboard provides an interactive analytical view of lifestyle factors affecting mental wellbeing.

5. DATASET OVERVIEW

- **Source:** Kaggle – Mental Health and Wellbeing Dataset
- **Target Variable:** Wellbeing_Category

Key Features:

- Hours of Sleep
- Stress Level
- Screen Time
- Physical Activity
- Social Interaction
- Work Hours
- Age
- Mental Health Status

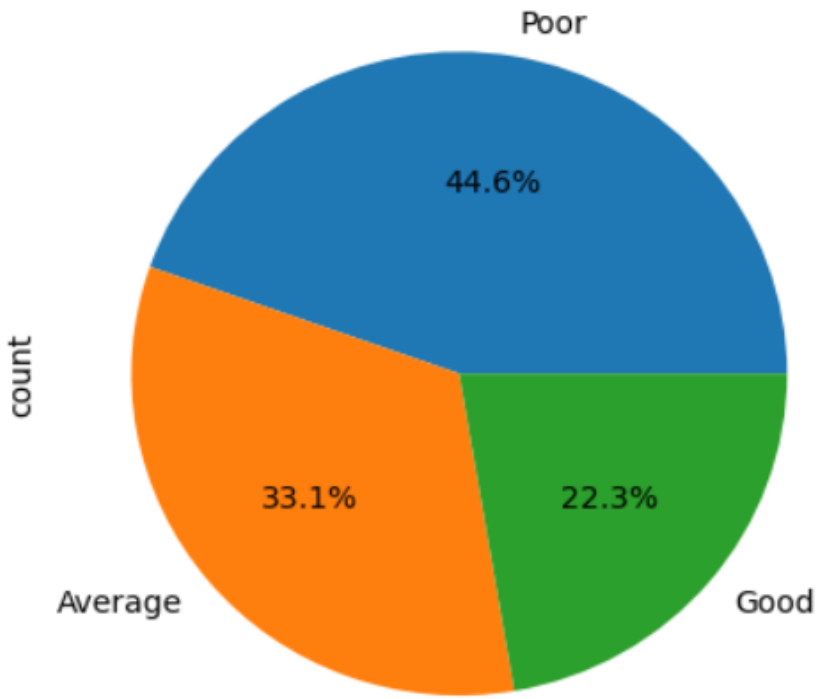
6. DATA ANALYSIS & EDA VISUALS

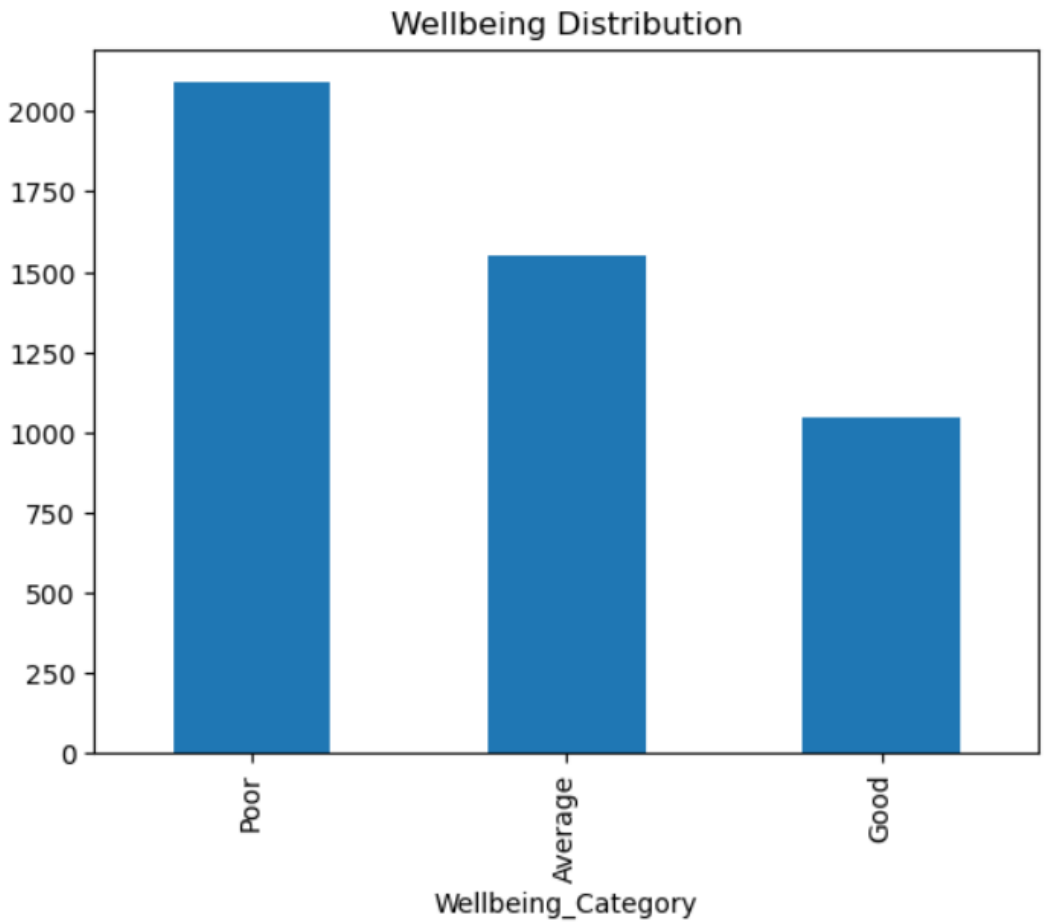
Key observations

- Adequate sleep improves wellbeing
- Higher stress leads to poorer mental health
- Excessive screen time negatively affects wellbeing.
- Physical activity improves mental health.
- Social interaction supports mental wellbeing.

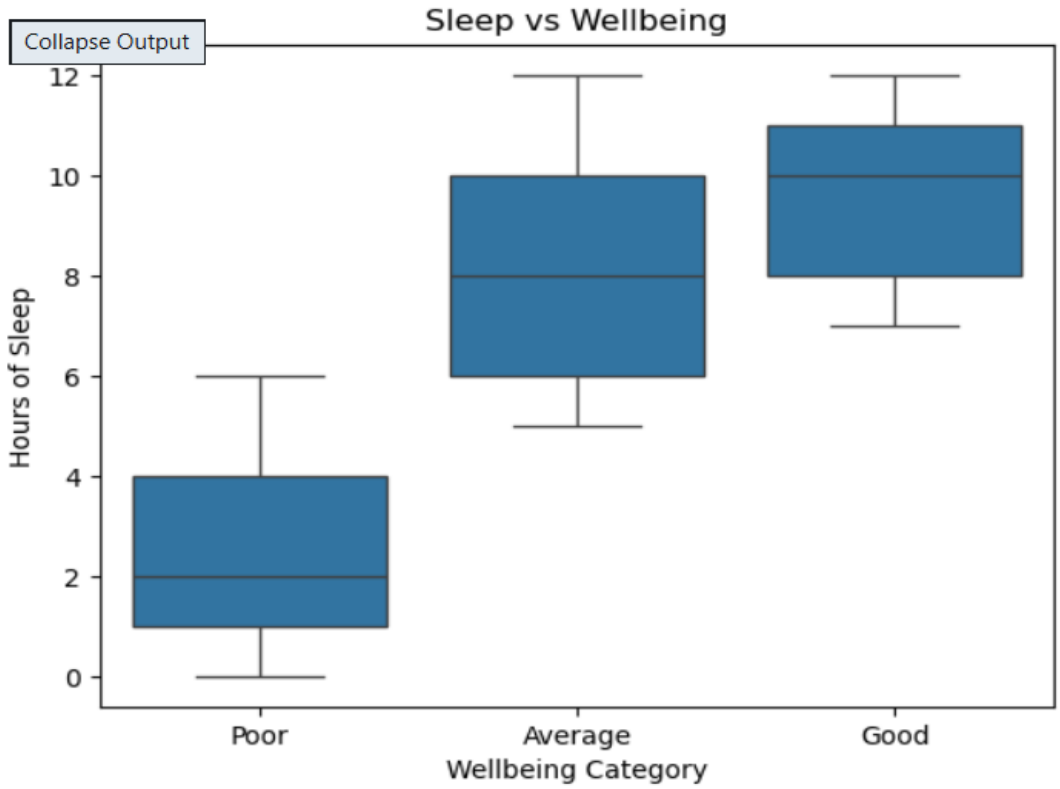
Visualizations Used:

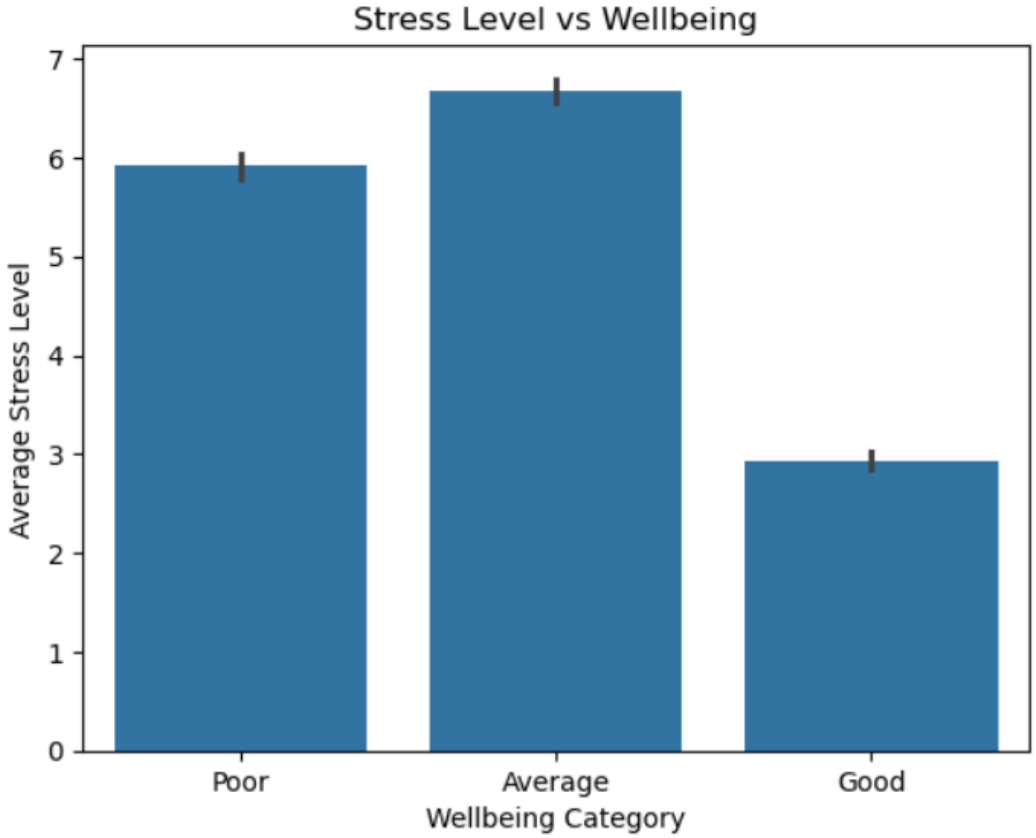
- Box plot
- Bar plot
- Pie chart
- Heat Map
- Model Accuracy
- Confusion Matrix

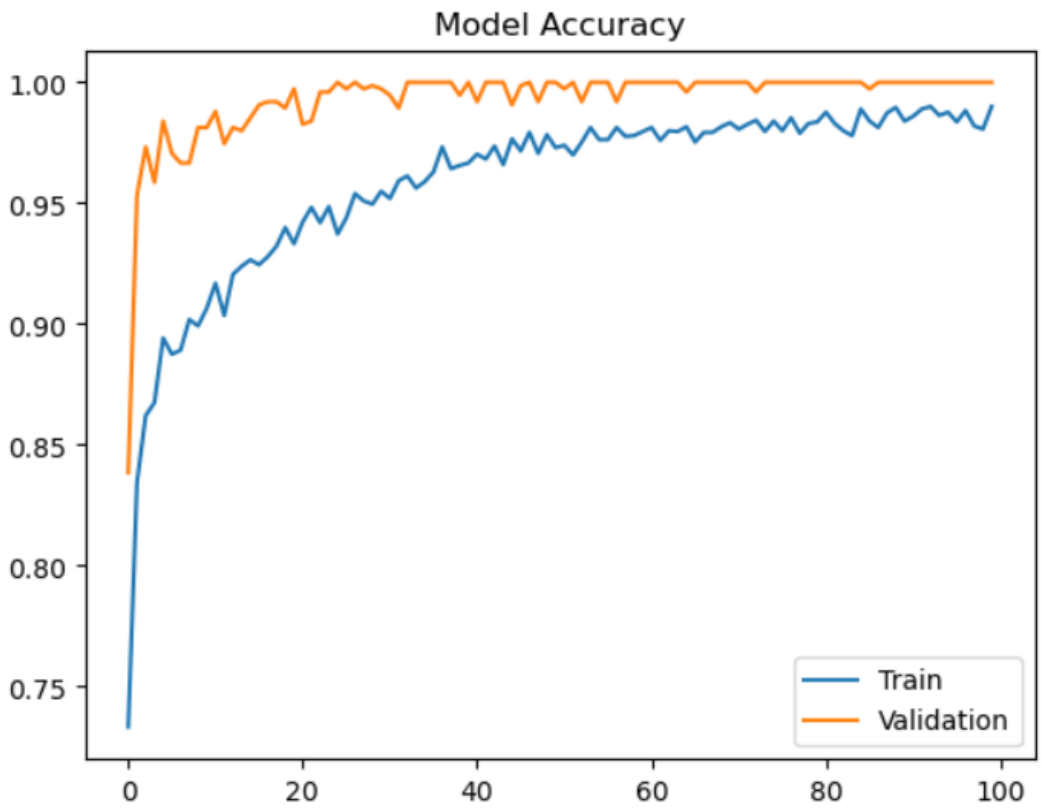

Poor
44.6%
count
33.1%
22.3%
Average
Good

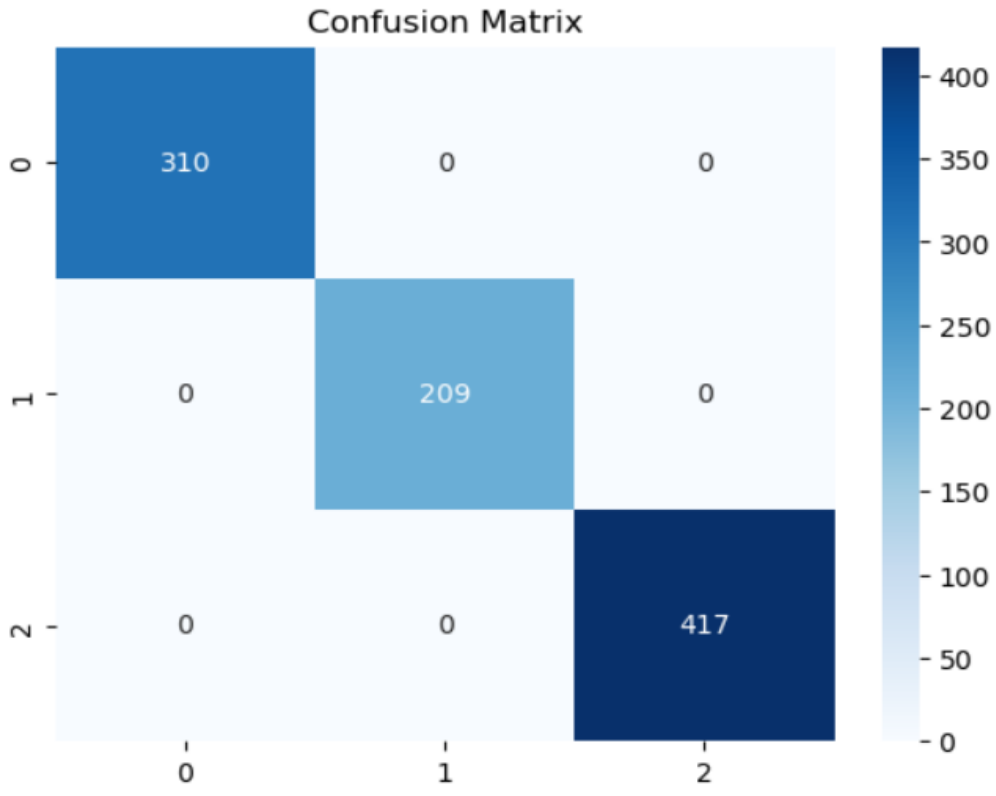

Wellbeing Distribution
2000
1750
1500
1250
1000
750
500
250
0
Poor
Average
Good
Wellbeing_Category

Collapse Output

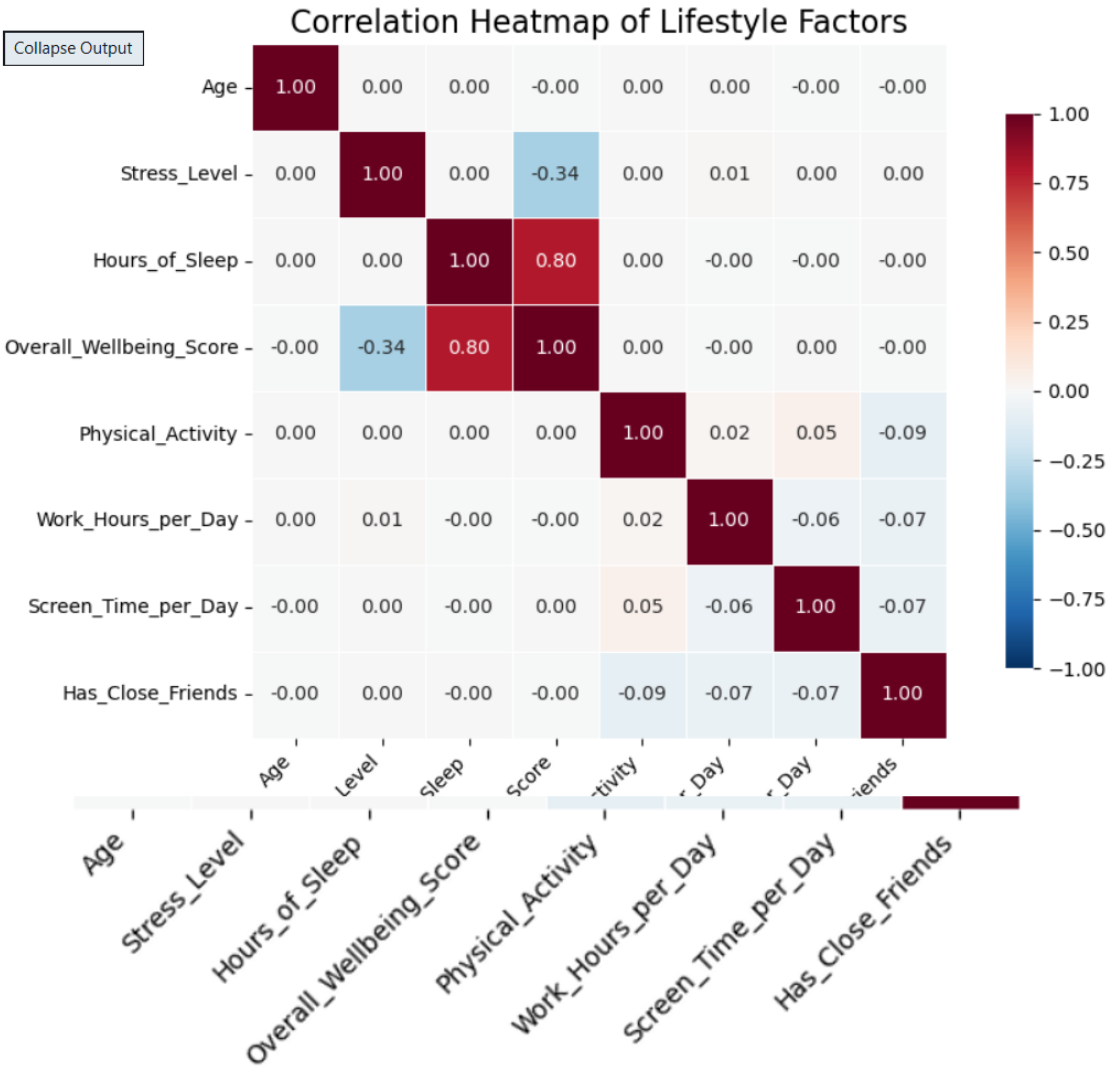





Model Accuracy
1.00
0.95
0.90
0.85
0.80
0.75
0
20
40
60
80
100
Train
Validation


Confusion Matrix
310
0
0
0
209
0
0
0
417
0
1
2
400
350
300
250
200
150
100
50
0

Collapse Output



7. MODEL DEVELOPMENT & RESULTS

Model Applied:

An Artificial Neural Network regression model was developed using TensorFlow/Keras

- **Model Architecture:**

  ~Input Layer – Lifestyle features

  ~ Dense Hidden Layer (ReLU)

  ~ Batch Normalization Layer

  ~ Dropout Layer – Overfitting prevention

  ~ Output Layer Softmax activation for classification

- **Compilation:**

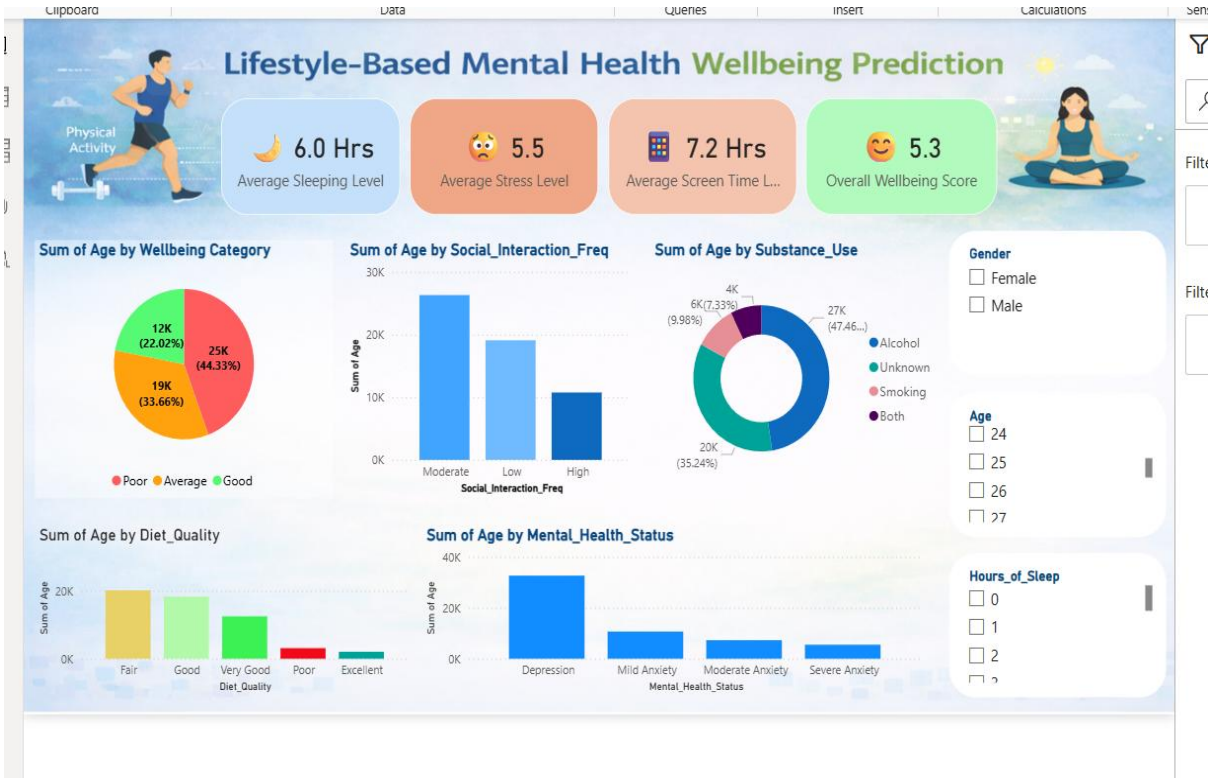
  ~ Optimizer → Adam

  ~ Loss Function → Categorical Crossentropy

  ~ Evaluation Metric → Accuracy

  ~ Epochs

8. DASHBOARD OVERVIEW



The Power BI dashboard titled "Lifestyle-Based Mental Health Wellbeing Prediction" provides an interactive analytical view of lifestyle behaviours and their impact on mental wellbeing.

KPIs Displayed:

- Average Sleeping Level
- Average Stress Level
- Average Screen Time
- Overall Wellbeing Score

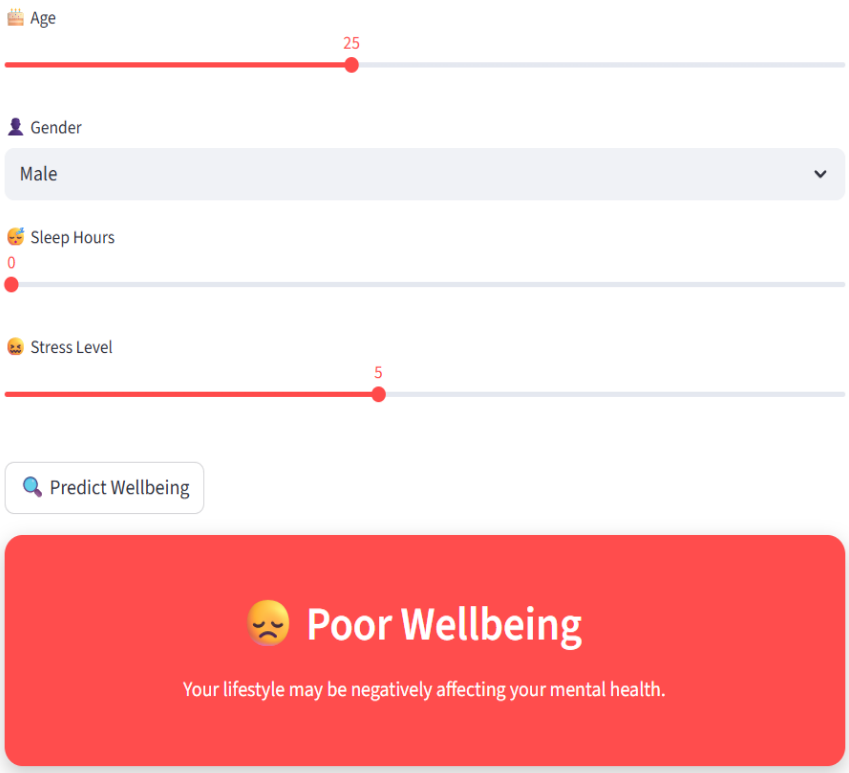
Visuals:

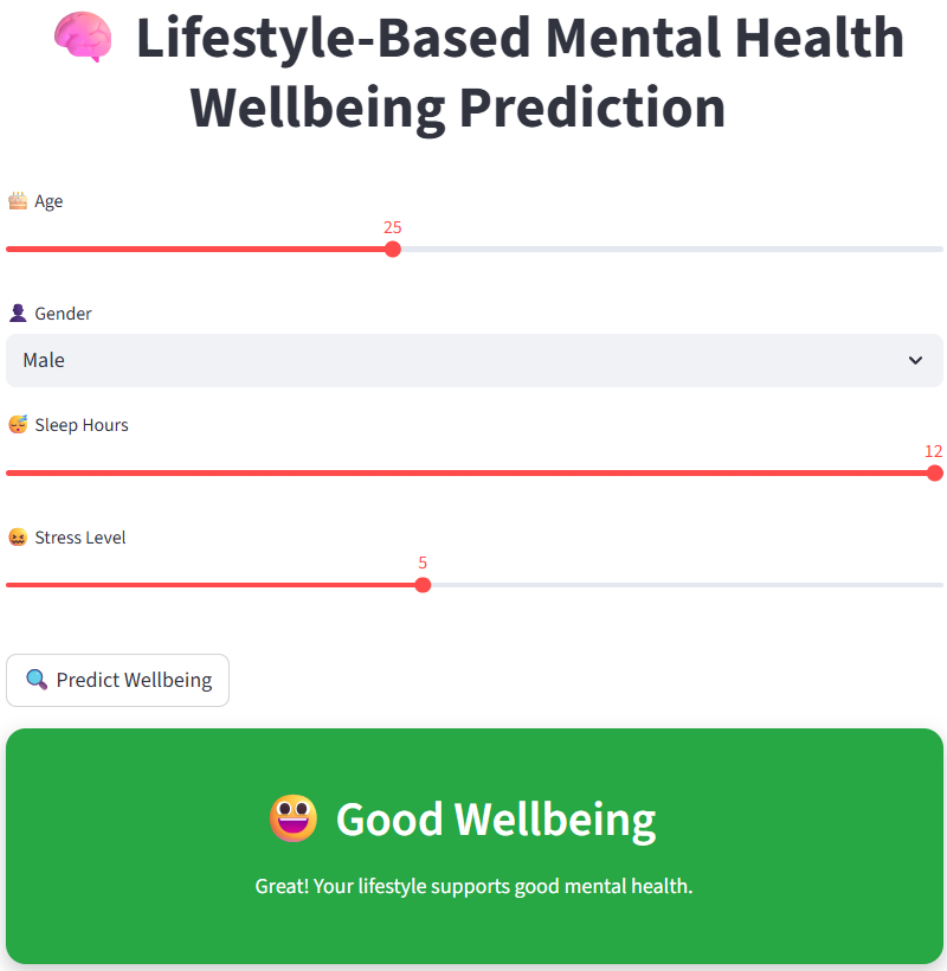
- Wellbeing Category Distribution (Pie Chart)
- Social Interaction Frequency Analysis (Bar Chart)
- Substance Use Distribution (Donut Chart)
- Diet Quality Analysis (Bar Chart)
- Mental Health Status Distribution (Column Chart)

Slicers:

- Gender
- Age
- Hours of Sleep

9. STREAMLIT WEB APP DEPLOYMENT





A Streamlit web application was developed to enable real-time mental wellbeing prediction based on user-input lifestyle factors such as age, gender, sleep hours, and stress level

10. CONCLUSION & FUTURE SCOPE

Conclusion:

The deep learning model successfully predicts mental wellbeing levels using lifestyle data. The project demonstrates how machine learning techniques can analyze behavioral patterns and provide insights into mental health.

This system can assist researchers and healthcare professionals in identifying lifestyle factors that influence mental wellbeing.

Future Scope:

- Integrate larger real-world mental health datasets
- Incorporate wearable device health data
- Develop mobile health monitoring applications
- Use advanced deep learning models such as LSTM
- Deploy the system as a real-time mental health prediction platform